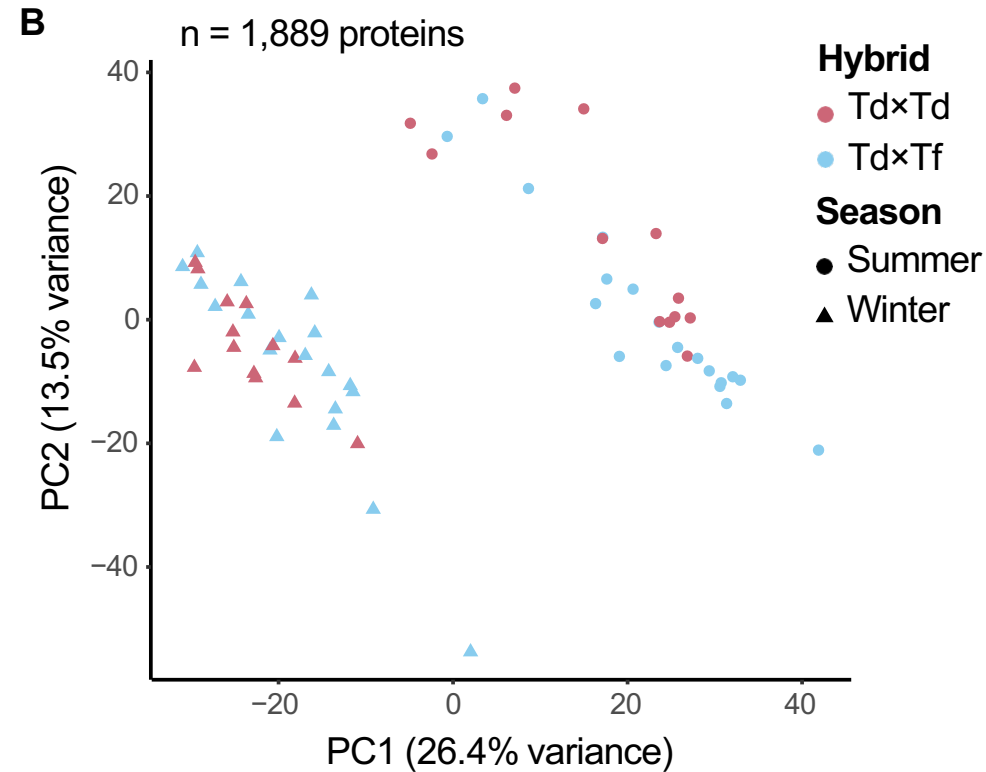
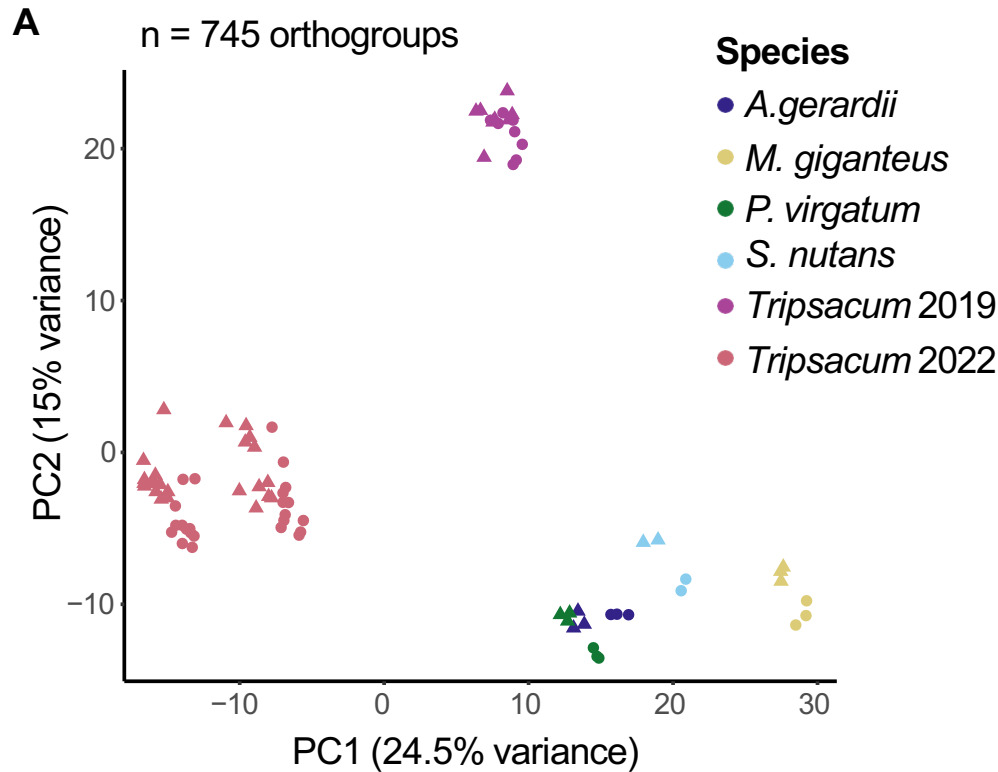
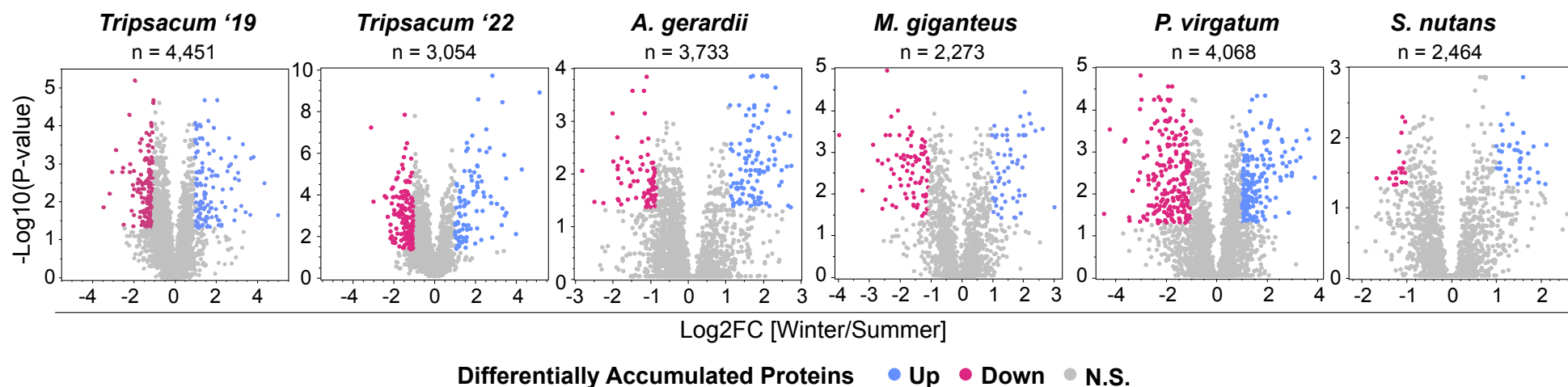
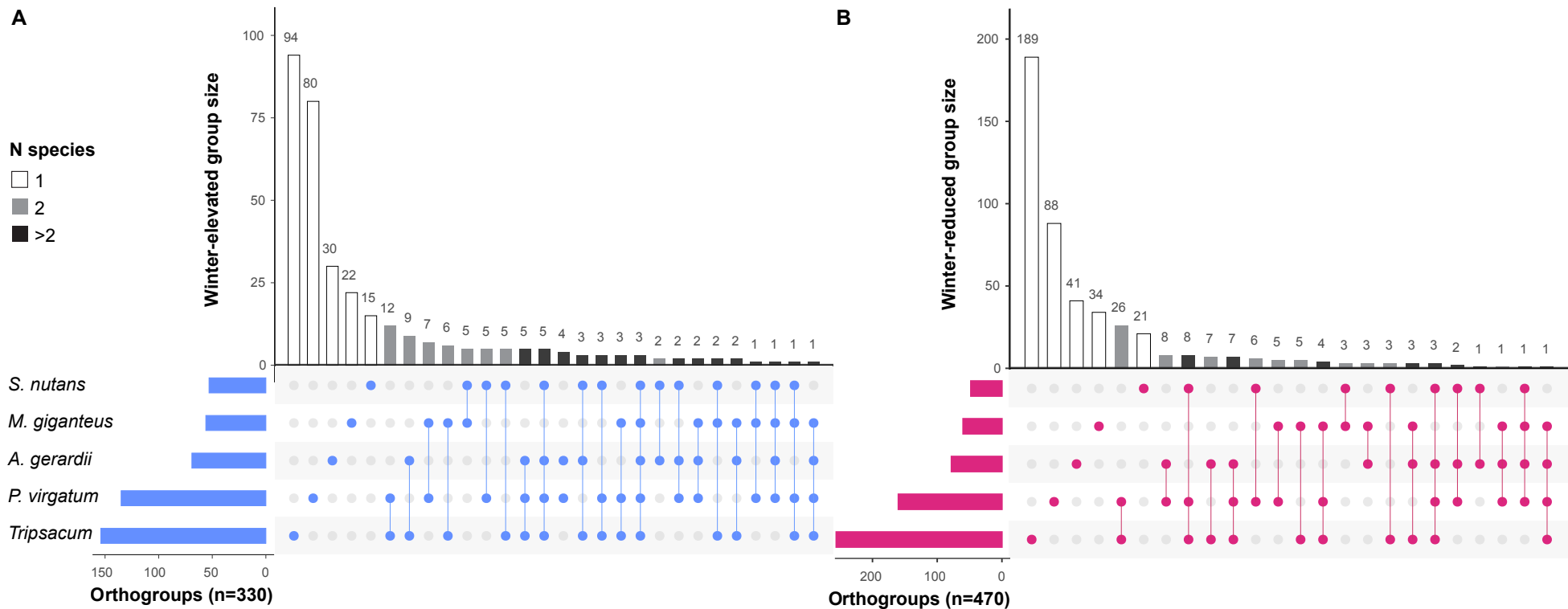




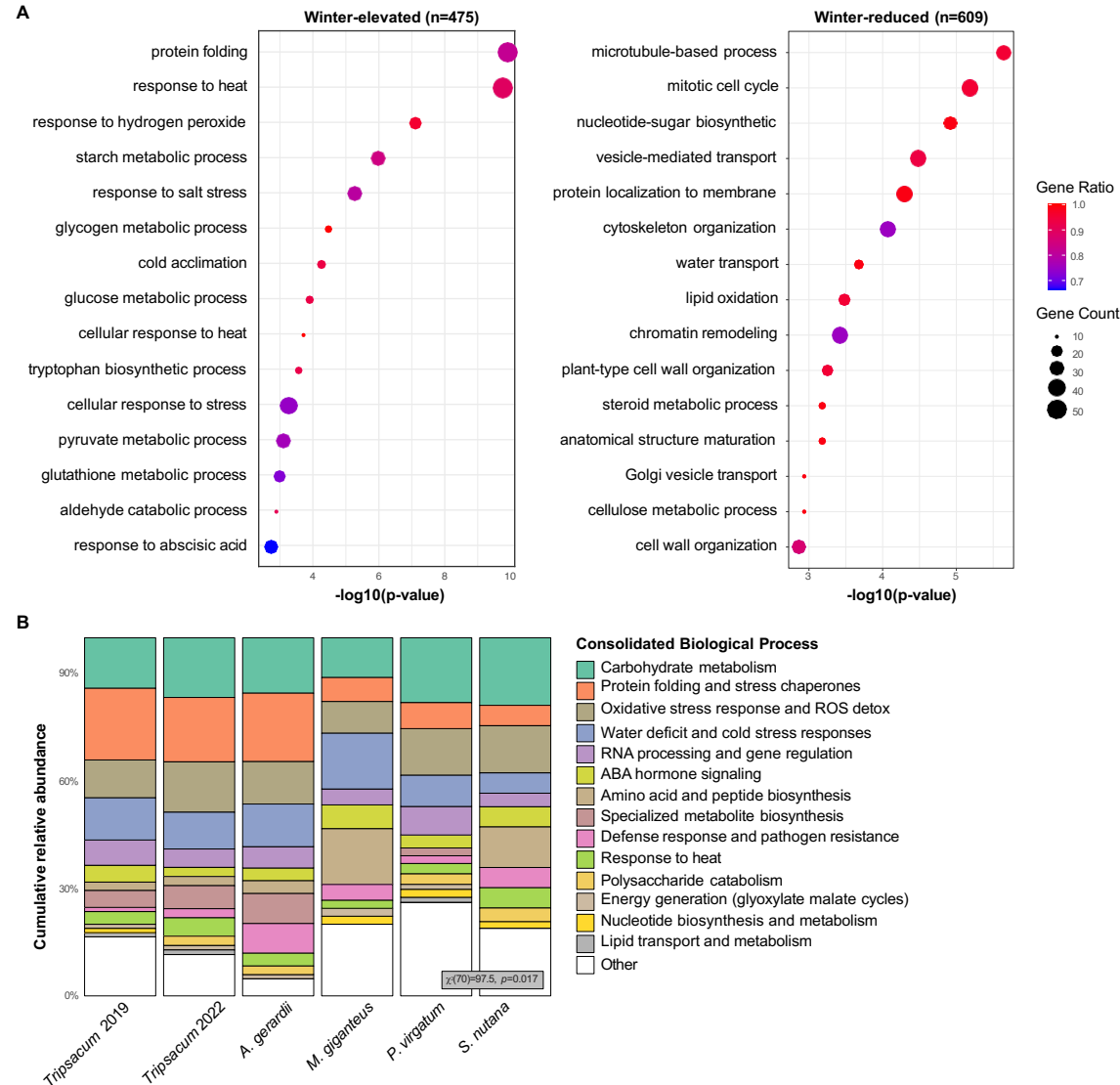
Supplemental Figure S1. Principal component analysis (PCA) of log-transformed protein abundance data. (A) PCA of 745 orthogroups from five species *Tripsacum* (Td-2019, Td-2022), *A. gerardii* (Ag), *M. giganteus* (Mg), *P. virgatum* (Pv), and *S. nutans* (Sn). **(B)** PCA of 1,889 proteins from *Tripsacum* hybrids. Points are colored by hybrid type—intraspecific *T. dactyloides* hybrids (Td × Td) and interspecific *T. dactyloides* × *T. floridanum* hybrids (Td × Tf). Shapes represent season (circle = summer, triangle = winter). The percentage of total variance explained by each principal component is shown on the axes.



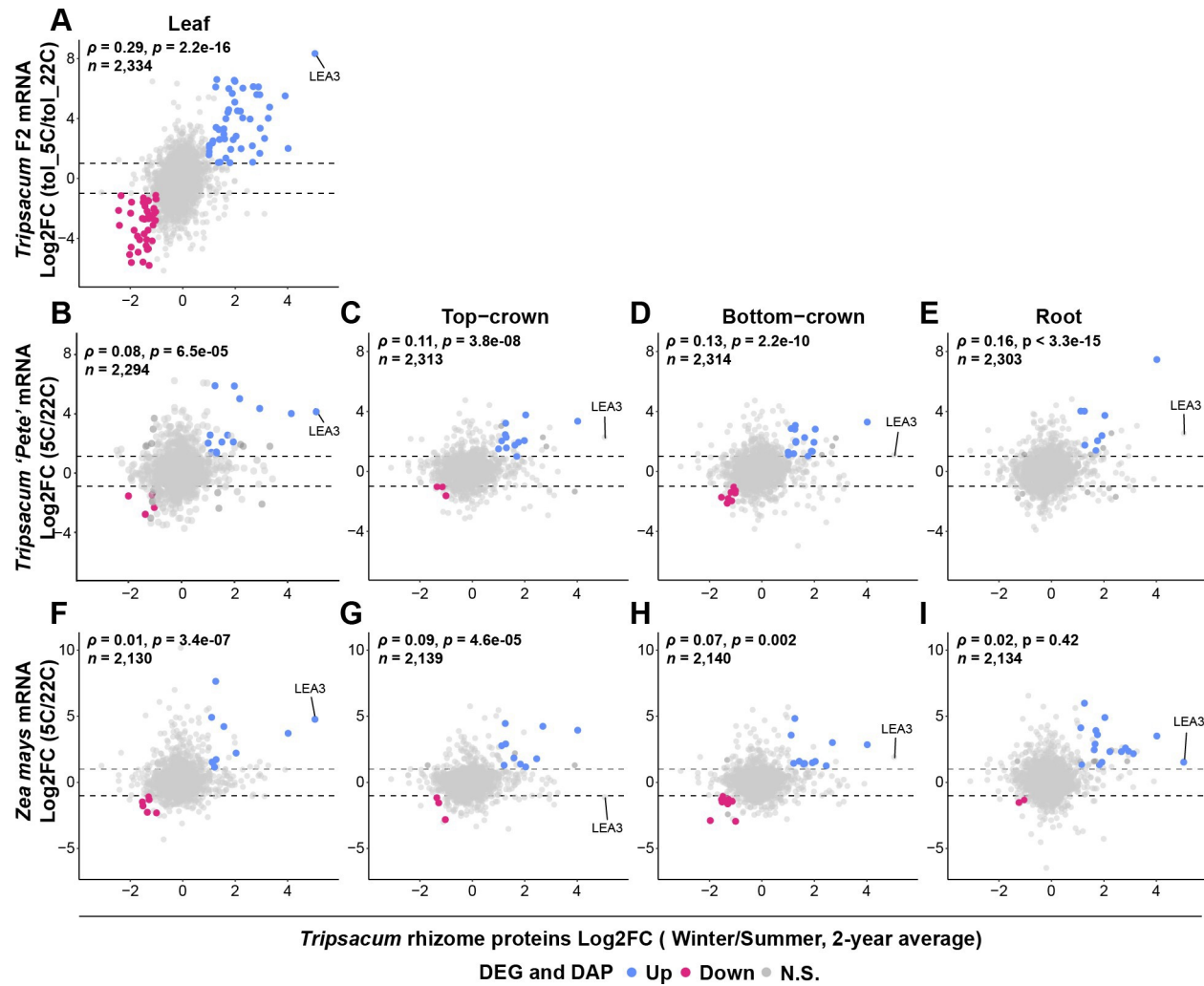
Supplemental Figure S2. Volcano plots showing differentially accumulated proteins (DAPs) in rhizomes of *Tripsacum* spp., *Andropogon gerardii*, *Miscanthus giganteus*, *Panicum virgatum*, and *Sorghastrum nutans* under winter vs. summer conditions, with total number of proteins identified listed. Proteins with significantly elevated (blue) and reduced (pink) abundance in winter relative to summer are highlighted, with non-significant proteins in gray. The y-axis represents the significance as $-\log_{10}(\text{p-value})$, and the x-axis shows the $\log_2$ fold-change ($\log_2\text{FC}$) of the DAPs between winter and summer.



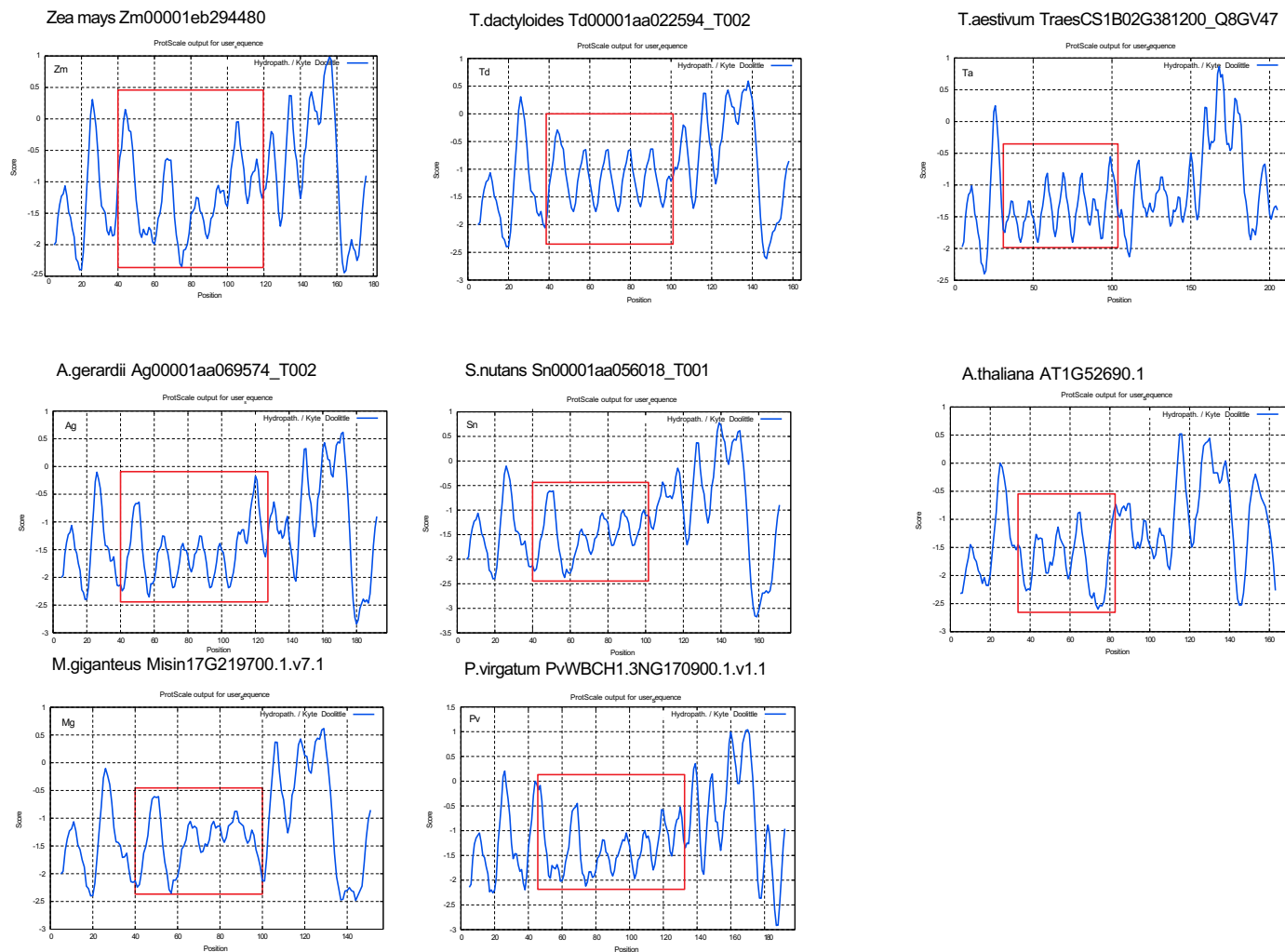
Supplemental Figure S3. Orthogroup sharing of significantly DAPs among five PACMAD grass species. (A) Winter-elevated. (B) Winter-reduced. Set sizes (horizontal bars) indicate the total number of differentially abundant orthogroups identified per species. Intersection sizes (vertical bars) represent orthogroups unique to individual species or shared among multiple species. Bar shading indicates the number of species sharing each orthogroup: unique to one species (white), shared between two species (gray), or shared among more than two species (black). Dots below the plot illustrate species-specific orthogroup intersections.



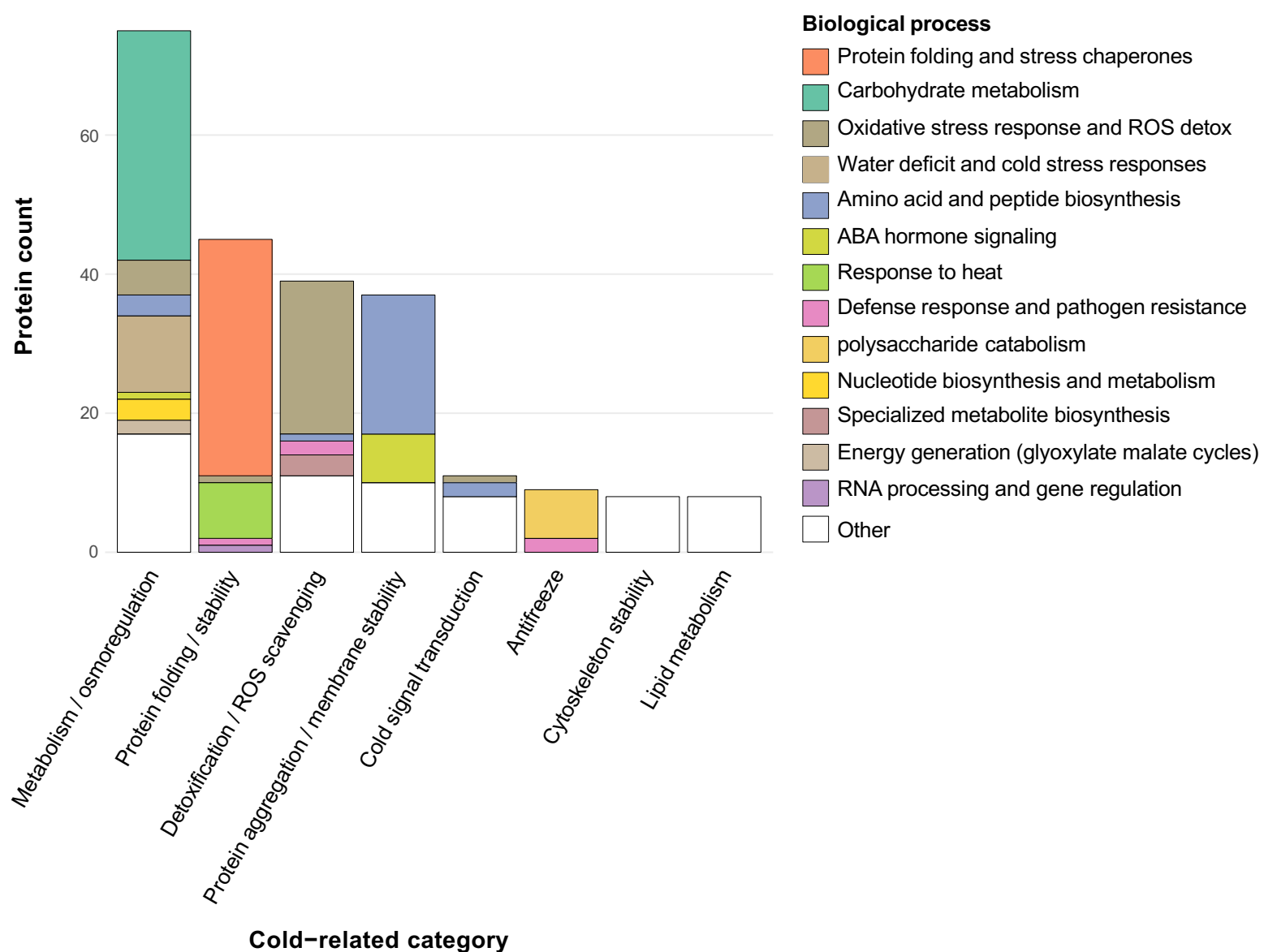
Supplemental Figure S4. (A) Enriched biological processes (BP) among differentially accumulated proteins (DAPs) in 475 winter-elevated and 609 winter-reduced proteins (proteins with unknown BP term were excluded). Dot size represents gene count per BP category, while dot color indicates the gene ratio (proportion of proteins annotated to each BP category). Only the top significantly enriched BP terms (lowest p-values) are displayed for clarity. **(B)** Mosaic plot of biological processes enriched among winter-elevated DAPs across five Andropogoneae species. The plot summarizes the distribution of functionally annotated proteins (n = 483 of 630; unannotated proteins were excluded) across 139 biological process (BP) terms. These terms were consolidated into broader functional categories to aid interpretability. Each bar shows the relative abundance of functional categories within a species, highlighting both conserved and species-specific patterns. A Chi-square test for independence ($\chi^2[70]=97.5, p=0.017$) was performed to evaluate variation across species. Category labels are shown in the legend.



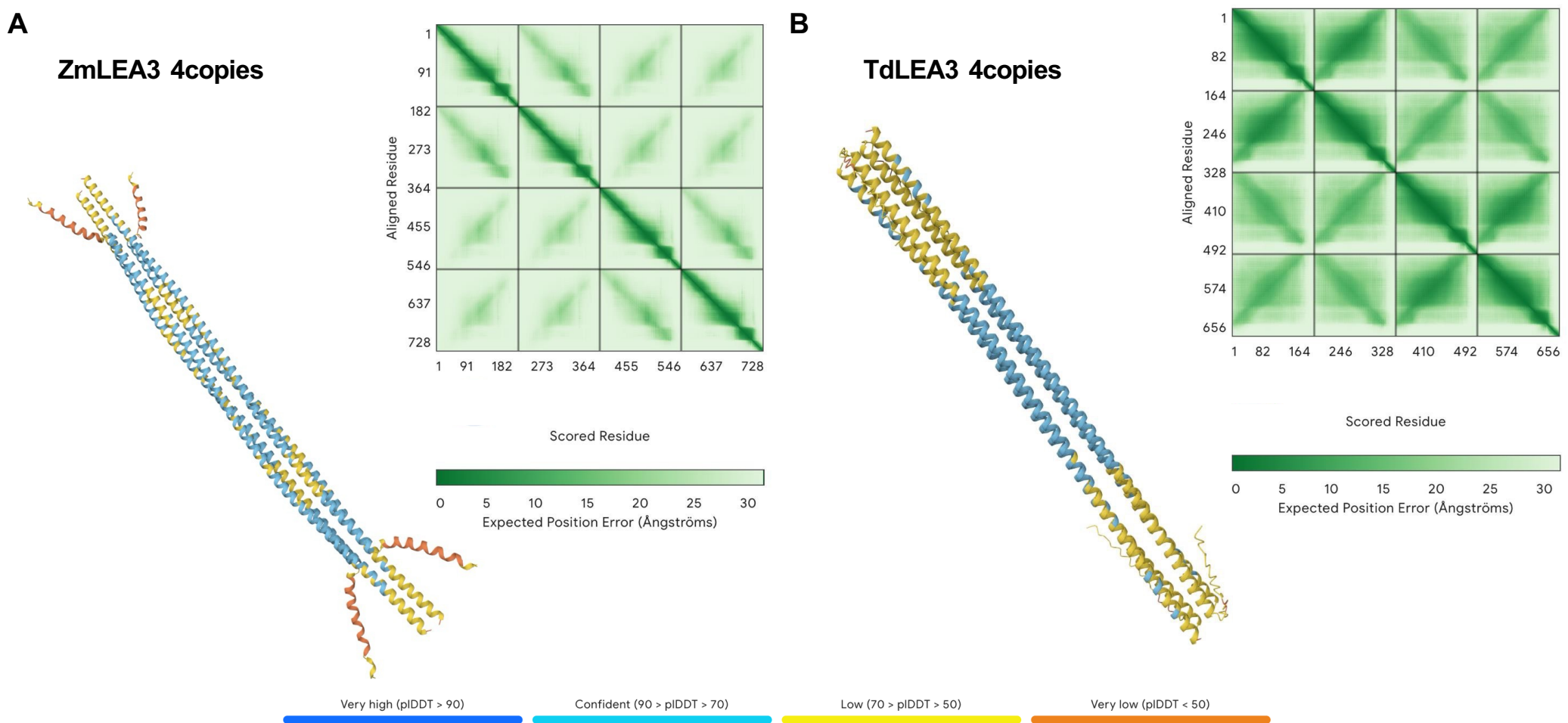
Supplemental Figure S5. Scatterplots showing the relationship between *Tripsacum* rhizome proteins (x-axis, log2FC Winter/Summer, 2-year average) and cold-responsive RNA expression in different seedling tissues (y-axis). (A) *Tripsacum* leaf mRNA from a freezing tolerant F bulk before and after 7 days cold acclimation (tol_5°C/tol_22°C) descended from the same *Tripsacum* hybrids used for rhizome proteomic analysis. (B-E) *Tripsacum* 'Pete' tissues (5°C/22°C): leaf, top-crown, bottom-crown, and root. (F-I) *Zea mays* tissues (5°C/22°C): leaf, top-crown, bottom-crown, and root. Points are colored by shared differential expression status: blue for upregulated, pink for downregulated, and gray for non-significant changes. Dashed horizontal lines indicate the ± 1 log2FC thresholds. Spearman's correlation coefficient (ρ) and p-value are shown in the upper left of each panel, along with the number of genes analyzed (n). The LEA3 gene is labeled in each panel.



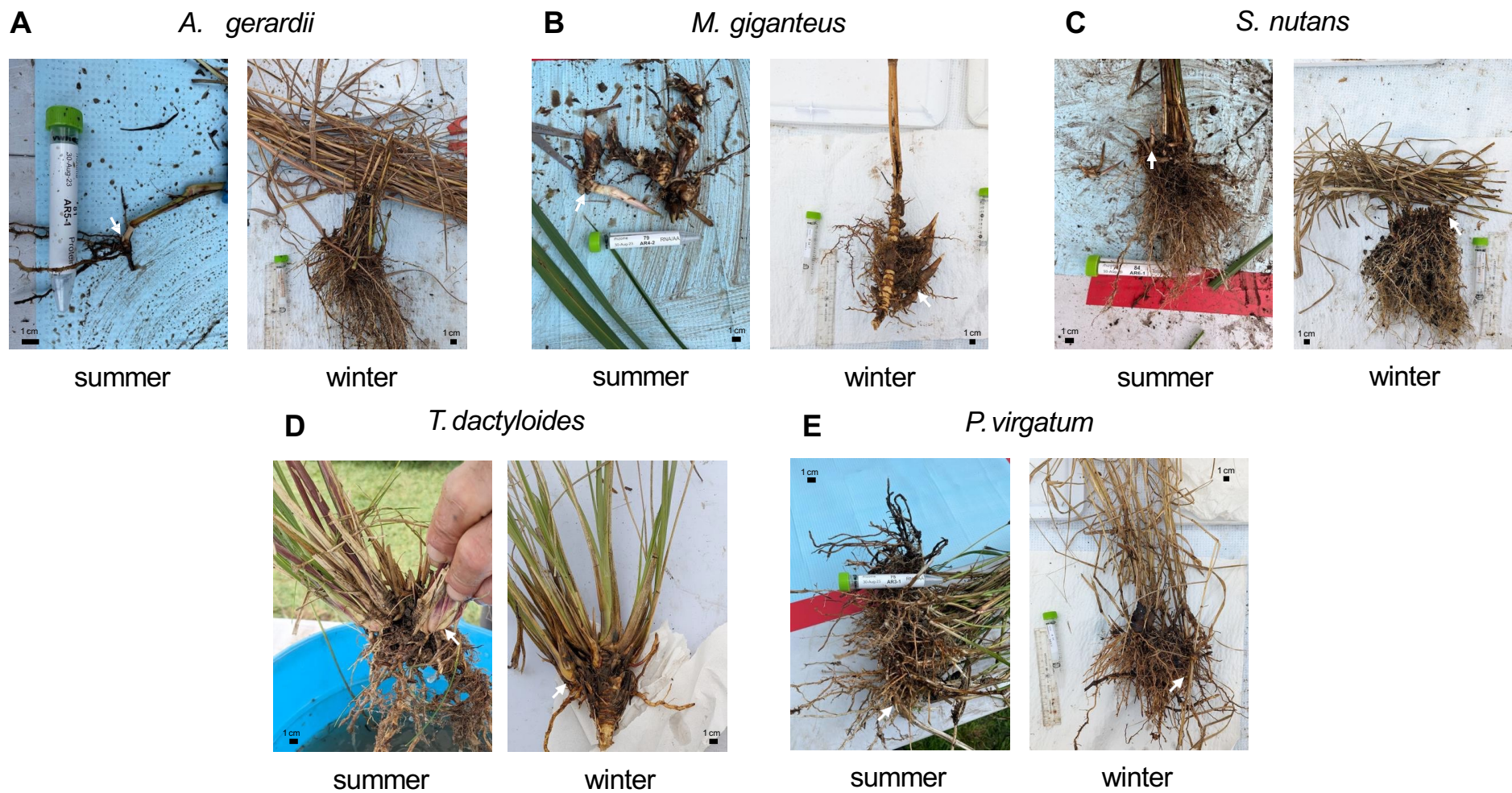
Supplemental Figure S6. Hydropathy profiles of LEA3 orthologs from eight species. Kyte-Doolittle plots were generated using ProtScale (ExPASy) with a window size of 11 amino acids. The x-axis indicates amino acid position, and the y-axis represents hydropathy score. Red boxes highlight regions corresponding to conserved 11-mer LEA motifs.



Supplemental Figure S7. Stacked bar chart illustrating the composition of BPs within curated cold-related functional categories. The plot includes 230 proteins from the top 50 DAPs accumulated in winter across species, after excluding those assigned to the cold response category “Other” (i.e., proteins with annotations deemed less relevant to cold tolerance interpretation). Cold-related categories are shown on the x-axis and sorted by total protein count. Each bar is color-coded by BP, highlighting the functional composition of each category. Less informative or poorly annotated BPs were grouped under “Other” in the legend. This visualization supports the manual consolidation of specific biological processes into broader cold-response categories used throughout the study.



Supplemental Figure S8. AlphaFold 3 prediction of LEA3 homo-tetramer assemblies. Predicted quaternary structures of LEA3 homotetramers for (A) *Zea mays* and (B) *Tripsacum dactyloides*. Both orthologs are predicted to form tetrameric assemblies. Colors indicate per-residue confidence (pLDDT), with dark blue/blue representing high to very high confidence. The *Z. mays* complex displays a distinct oligomeric packing arrangement relative to *Tripsacum*. Predicted Aligned Error (PAE) plots. Low error values (dark green) in the off-diagonal regions confirm high confidence in the relative positioning of the subunits, validating the predicted tetrameric interfaces for both species.



Supplemental Figure S9. Rhizome morphology at sampling. Paired images display representative rhizome tissues (white arrows) collected during summer (August, left) and winter (January, right) for **(A)** *Andropogon gerardii*, **(B)** *Miscanthus giganteus*, **(C)** *Sorghastrum nutans*, **(D)** *Tripsacum dactyloides*, **(E)** and *Panicum virgatum*. Summer samples exhibit active bud growth and pale storage tissue. Winter samples exhibit dormancy and epidermal hardening. Proteomic analysis was restricted to mid-section of these storage organs. Scale bars represent approx. 1 cm.